1.

Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Comparison Table: DDR3 vs DDR4 vs DDR5 RAM

RAM (Random Access Memory) is a temporary memory that stores the data and instructions a computer needs while running programs. DDR3, DDR4, and DDR5 are different generations of RAM, with each newer generation generally offering higher speed and better efficiency.

Feature	DDR3 RAM	DDR4 RAM	DDR5 RAM
Full Form	Double Data Rate 3	Double Data Rate 4	Double Data Rate 5
Data Transfer Speed	800–2133 MT/s	1600–3200 MT/s	4800–8800+ MT/s
Operating Voltage	1.5 V (typically)	1.2 V	1.1 V
Performance	Low compared to newer generations	Better than DDR3	Higher performance than DDR4
Power Consumption	Higher	Lower than DDR3	Lower operating voltage and improved power efficiency
Typical Capacity per Module	Commonly 2–8 GB	Commonly 4–32 GB	Commonly 8–64 GB or more
Typical Use Cases	Older desktop computers and laptops	Many older and mid-range desktop computers and laptops	Modern desktop computers, gaming PCs, and high-performance systems
Compatibility	Requires a DDR3-compatible motherboard	Requires a DDR4-compatible motherboard	Requires a DDR5-compatible motherboard

2.

Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

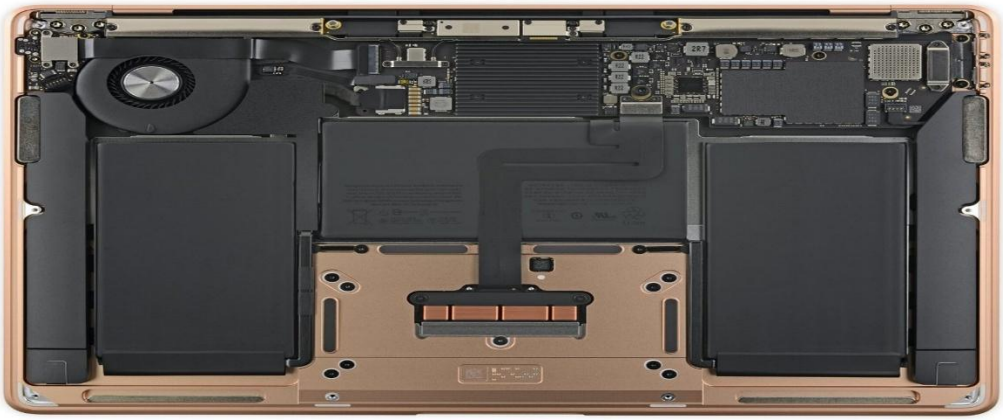
Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

Aim

To locate the RAM slots, identify the installed RAM module, safely reinstall it, and verify that the computer starts successfully.

Procedure

1. **Shut down the computer:** Turn off the laptop or desktop completely and disconnect the power supply. For a desktop, unplug the power cable. For a laptop, disconnect the charger and follow the manufacturer's instructions for disconnecting the battery.
2. **Locate the RAM slots:** Open the computer's case or the laptop's designated RAM access panel according to the manufacturer's manual.
3. **Identify the RAM module:** Locate the installed RAM module in its slot. Observe its label and note its capacity and type, such as DDR4 o



3.

List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

HDD, SSD, and NVMe are storage technologies used to store operating systems, applications, and personal files. They differ in speed, technology, and performance.

Comparison Table

Feature	HDD (Hard Disk Drive)	SSD (Solid-State Drive)	NVMe (Non-Volatile Memory Express)
1. Speed	Slow; typically 80–160 MB/s	Fast; typically 400–550 MB/s for SATA SSDs	Very fast; commonly 3,000–7,000 MB/s for PCIe Gen 3 and Gen 4 drives
2. Technology	Uses spinning magnetic disks	Uses flash memory with no moving parts	Uses flash memory and the NVMe protocol, usually over PCIe
3. Durability	More sensitive to physical shocks because it has moving parts	More resistant to physical shocks	More resistant to physical shocks than HDDs because it has no moving parts

Feature	HDD (Hard Disk Drive)	SSD (Solid-State Drive)	NVMe (Non-Volatile Memory Express)
Best Use Case	Storing large files, photos, and videos	Running everyday applications and operating systems	Gaming, video editing, and running demanding applications
Example Application	Google Photos backups or storing downloaded movies	Microsoft Word, Chrome, and everyday applications	Adobe Premiere Pro or large PC games

4.Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

1. Aim

To identify the storage drive (HDD or SSD), recognize its connector type, understand the removal and reinstallation process, and verify that the computer detects the storage device after reassembly.

2. Types of Storage Connectors

Storage Type	Connector Type	Description
HDD	SATA	Uses a SATA data connector and a separate power connector in most desktop computers.
SATA SSD	SATA	Uses a SATA interface to connect to the computer.
M.2 SATA SSD	M.2 SATA	Fits into an M.2 slot and communicates using the SATA interface.
M.2 NVMe SSD	M.2 NVMe	Fits into an M.2 slot and uses PCIe for high-speed data transfer.

3. Procedure

1. Shut down the computer and disconnect the power cable or charger.
2. Follow the manufacturer's service manual to open the computer safely.
3. Locate the HDD or SSD inside the computer.
4. Identify the connector type (SATA or M.2).
5. If you are authorized and comfortable doing so, carefully remove the drive according to the manufacturer's instructions.
6. Reinstall the drive securely and close the computer.
7. Reconnect the power and turn on the computer.

8. Open **Task Manager → Performance → Disk** in Windows to check whether the storage device is detected.

Safety precaution: Do not force the drive or connector. Follow the manufacturer's instructions for disconnecting the battery and protecting the device from electrostatic discharge.

4. YouTube Video Reference

If you do not have access to a computer you are allowed to open, use a video demonstration.

Topic: Installing a SATA SSD in a laptop.

Watch the tutorial here:

Kingston SSD Installation Tutorial on YouTube

For the most accurate instructions, search YouTube for your specific laptop or desktop model.

5. Observation Table

Observation	Details
Storage Device	HDD / SSD
Connector Type	SATA / M.2 SATA / M.2 NVMe
Storage Capacity	_____ GB
Drive Reinstalled	Yes / No
System Boot Status	Successful / Unsuccessful
Storage Detected	Yes / No

6. Result

The storage drive and its connector types were studied. The removal and reinstallation process was reviewed, and the method of checking storage detection after reassembly was understood.